GridPulse: An Integrated Machine Learning System for Renewable Energy Forecasting and Carbon-Aware Battery Dispatch

Pratik Niroula
MNSU CIT (Major), Math (Minor)

February 19, 2026

Abstract

GridPulse is an end-to-end decision system that links forecasting, uncertainty estimation, battery dispatch optimization, operational safety controls, and post-decision evaluation for power-system operations. The implementation follows a closed loop (*Forecast* → *Optimize* → *Dispatch* → *Measure* → *Monitor*) so model outputs are translated into constrained actions and measured against baselines in the same artifact pipeline. This thesis uses a strict dataset-scoped latest evidence policy locked to artifacts dated February 17, 2026. Canonical decision-impact sources are `reports/impact_summary.csv` for Germany (DE) and `reports/eia930/impact_summary.csv` for the United States (US). Canonical stochastic metrics are sourced from `reports/research_metrics_de.csv` run `20260217_165756` and `reports/research_metrics_us.csv` run `20260217_182305`.

Under this lock, DE impact is 7.11% cost savings, 0.30% carbon reduction, and 6.13% peak shaving. US impact is 0.11% cost savings, 0.13% carbon reduction, and 0.00% peak shaving. Stochastic outcomes are DE $\text{EVPI}_{\text{robust}}$ 2.32, DE $\text{EVPI}_{\text{deterministic}}$ -30.40, DE VSS 2,708.61, and US $\text{EVPI}_{\text{robust}}$ 10,279,851.74, US $\text{EVPI}_{\text{deterministic}}$ 24,915,503.93, US VSS 297,092.71. Beyond these outcomes, the core contribution is governance: publication-facing claims are traceable to source files, run IDs, and rounding rules through `paper/metrics_manifest.json`, `paper/claim_matrix.csv`, and `scripts/validate_paper_claims.py`.

Keywords: Machine Learning, Energy Forecasting, Battery Dispatch, Robust Optimization, Conformal Prediction, Stochastic Programming, MLOps, Grid Operations

1 Introduction

1.1 Operational Motivation

Prediction-only systems do not deliver operational action. Grid operators need feasible dispatch decisions under uncertainty while balancing cost and carbon objectives. GridPulse is designed as a closed decision loop:

$$\text{Forecast} \rightarrow \text{Optimize} \rightarrow \text{Dispatch} \rightarrow \text{Measure} \rightarrow \text{Monitor}$$

1.2 Problem Statement

The central problem is integrated decision quality rather than isolated forecast error. The system must jointly satisfy five requirements: accurate short-horizon forecasting, uncertainty quantification usable in optimization, physically feasible dispatch, measurable impact vs. baselines, and reproducible claim traceability across manuscript formats.

1.3 Research Questions

1. Can one architecture maintain strong forecasting quality across DE and US datasets with different operating regimes?
2. Does uncertainty-aware dispatch provide measurable value relative to deterministic dispatch?
3. What is the magnitude of cost/carbon/peak impact under dataset-scoped latest evidence locking?
4. Can run-scoped governance prevent cross-document metric contradictions?

1.4 Contributions

1. A production-oriented multi-layer GridPulse system spanning data, forecasting, optimization, monitoring, and serving.
2. A conformal and adaptive interval workflow integrated with robust dispatch evaluation.
3. A dataset-scoped latest metric lock with explicit run IDs and source traceability.
4. A publication governance mechanism (manifest, claim matrix, validator) that blocks legacy metric regression.

Contribution boundaries are explicit: this thesis does not claim universal market realism across all settlement regimes or universal transferability across all operators. It claims reproducible DE/US decision outcomes under the locked artifacts and governance policy.

1.5 Thesis Scope and Reading Guide

This draft is structured so sections can be expanded or trimmed without breaking metric consistency:

1. Sections 2–4 define implementation and methods with code-level anchors.
2. Sections 5–8 define evidence policy, traceability, and claim governance.
3. Section 6 provides canonical quantitative results.
4. Sections 9–13 cover interpretation, validity, limitations, and roadmap.
5. Appendices provide replication commands, artifact inventory, and copy-ready writing guidance.

If length must be reduced, preserve Sections 6, 7, and 14 first because they carry the metric-locked publication claims.

2 System Architecture and Implementation

2.1 Decision Loop and Layering

GridPulse follows an eight-layer decision architecture implemented across repository modules:

1. Data ingestion and feature generation.
2. Forecast model training and inference.
3. Uncertainty interval calibration.

4. Anomaly detection.
5. Dispatch optimization (deterministic and robust).
6. Impact and stochastic evaluation.
7. Monitoring and retraining decisions.
8. API and dashboard serving.

2.2 Component-to-Code Map

Table 1: Architecture Components and Primary Code Locations

Layer	Purpose	Primary paths
Data pipeline	Feature generation and split construction	`src/gridpulse/data_pipeline/`, `src/gridpulse/pipeline/run.py`
Forecasting	GBM/LSTM/TCN training and inference	`src/gridpulse/forecasting/`
Uncertainty	Conformal + adaptive interval logic	`src/gridpulse/forecasting/uncertainty/conformal.py`
Optimization	Deterministic and robust dispatch solvers	`src/gridpulse/optimizer/lp_dispatch.py`, `src/gridpulse/optimizer/robust_dispatch.py`
Monitoring	Drift tests and retraining decisions	`src/gridpulse/monitoring/`
Safety	BMS checks and watchdog controls	`src/gridpulse/safety/`
API serving	API endpoint routing	`services/api/main.py`, `services/api/routers/`
Frontend	Operator dashboard and region views	`frontend/`

2.3 Runtime Interfaces

FastAPI serving endpoints include:

- `/forecast`
- `/forecast/with-intervals`
- `/anomaly`
- `/optimize`
- `/monitor`
- `/monitor/model-info`
- `/dc3s/step`
- `/dc3s/audit/\{command\_id\}`
- `/health`, `/ready`, `/metrics`

Additional control endpoints include `/system/health`, `/system/heartbeat`, and `/control/dispatch`, gated by API-key scope and safety validation. Existing runtime endpoints remain unchanged; these interfaces are additive extensions for Section 1–3 implementation coverage.

2.4 Artifact Handoff Contracts

This thesis uses the following artifact families:

1. Dataset profile and dashboard metrics: `data/dashboard/*.json`.
2. DE impact summary: `reports/impact_summary.csv`.
3. US impact summary: `reports/eia930/impact_summary.csv`.
4. Stochastic run metrics: `reports/research_metrics_de.csv` and `reports/research_metrics_us.csv`.
5. Robustness diagnostics: `reports/publication/tables/table6_robustness.csv`.

These artifact contracts are treated as publication interfaces. If a contract path or schema changes, both `paper/metrics_manifest.json` and `paper/claim_matrix.csv` must be updated in the same revision.

2.5 Deployment and Runtime Modes

GridPulse supports three execution modes:

1. Offline batch mode for training, evaluation, and report generation.
2. API service mode (`services/api/main.py`) for live forecast and optimization requests.
3. Dashboard mode (`frontend/`) for operator-facing visual workflows.

The system is documented with safe degradation paths (for example, fallback baseline dispatch on infeasible optimization states) rather than assuming uninterrupted perfect runtime conditions.

2.6 End-to-End Request Lifecycle

An operational request path is:

1. Load latest feature window and resolve requested targets.
2. Retrieve model bundle and generate forecast vectors.
3. Attach conformal intervals when calibration artifacts exist.
4. Solve deterministic or robust dispatch with configured constraints.
5. Return action plan and cost/carbon summaries when available.
6. Emit monitoring signals and preserve control-plane heartbeat state.

2.7 Section 1–3 Implementation Update (Publish Blockers, Streaming, DC3S)

Section 1 (immediate publish blockers) was implemented by replacing frontend chat-tool mock payloads with backend-derived FastAPI calls in `frontend/src/app/api/chat/tool-executors.ts`. Production tool paths now resolve through `/forecast`, `/forecast/with-intervals`, `/optimize`, `/optimize/baseline`, `/monitor`, `/anomaly`, `/health`, `/ready`, and `/monitor/model-info`. Tool execution explicitly fails on backend errors instead of returning fabricated fallback payloads. Deployment/runtime hygiene updates were applied as well: dashboard ports were normalized to `3000`, dashboard deploy workflow Dockerfile targeting was corrected, local absolute report paths were removed, and coverage artifacts were removed/ignored for reproducibility hygiene.

Section 2 (streaming correctness) was implemented with a new CLI module `src/gridpulse/streaming/run_consumer.py` supporting: `python -m gridpulse.streaming.run_consumer -config configs/streaming.yaml -max-messages N`. This loader maps YAML config into streaming `AppConfig`, instantiates `StreamingIngestConsumer`, and closes resources cleanly. Persistence behavior was made explicit via public `write_event(...)` in `src/gridpulse/streaming/consumer.py` and integrated in `src/gridpulse/streaming/worker.py` after schema validation. The expected persistence artifact is `data/interim/streaming.duckdb` with table `telemetry_events`.

Section 3 (DC3S) was implemented with new config `configs/dc3s.yaml`, DC3S modules in `src/gridpulse/dc3s/` (`quality.py`, `drift.py`, `calibration.py`, `shield.py`, `certificate.py`, `state.py`), and FastAPI router endpoints in `services/api/routers/dc3s.py`: `POST/dc3s/step` and `GET/dc3s/audit/\{command\_id\}`. The step endpoint returns proposed and shield-repaired safe actions, uncertainty payloads, and certificate identifiers; audit retrieval returns persisted certificate JSON. The `heuristic` controller mode is intentionally not implemented in this phase and is explicitly rejected at runtime.

Frontend live-ops behavior for this flow is explicitly implemented in `frontend/src/app/(dashboard)/page.tsx`, `frontend/src/lib/api/dc3s-client.ts`, `frontend/src/components/dashboard/DC3SLiveCard.tsx`, and `frontend/src/app/api/dc3s/audit/[commandId]/route.ts`. The dashboard initializes DC3S polling at 15 seconds, supports operator-selected auto-refresh (`Off`, `5s`, `10s`, `15s`, `30s`, `60s`), and provides manual refresh. The live card exposes a command-linked audit route (`/api/dc3s/audit/\{commandId\}`) that proxies to FastAPI `/dc3s/audit/\{commandId\}` with explicit error propagation.

3 Data Assets, Scope, and Feature Design

3.1 Dataset Policy Lock (Canonical)

This manuscript follows the dashboard profile lock in `data/dashboard/manifest.json` generated at `2026-02-17T11:15:38.623283`.

Table 2: Canonical Dataset Profile Lock

Region	Rows (hourly)	Columns	Engineered Features	Approx Days
DE	17,377	98	94	724.04
US	13,638	118	114	568.25

Date ranges (UTC):

- DE: 2018-10-07 23:00 to 2020-09-30 23:00
- US: 2024-07-01 06:00 to 2026-01-20 11:00

3.2 Signals and Targets

Both regions use three target variables: `load_mw`, `wind_mw`, and `solar_mw`. The DE profile includes price/carbon channels specific to OPSD-based processing, while the US profile includes EIA-930 generation-mix channels and region-specific price/carbon features.

3.3 Feature Family Composition

3.4 Target Distribution Snapshot

3.5 Scope Boundary and Legacy Drift

Legacy profile claims from older report families are out of scope unless explicitly reconciled to this dashboard lock. This avoids reintroducing conflicting row/date/feature statements.

Table 3: Engineered Feature Family Counts

Region	Weather Features	Lag Features	Calendar Features	Total Features
DE	40	36	6	94
US	56	42	6	114

Table 4: Target Statistics from Dashboard Profile Files

Region	Target	Mean (MW)	Std (MW)	Min (MW)	Max (MW)	Median (MW)	Non-zero (%)
DE	`load_mw`	55,156.42	9,998.06	31,923.0	76,925.0	54,752.0	100.0
DE	`wind_mw`	14,368.98	10,321.58	136.0	46,064.0	11,728.0	100.0
DE	`solar_mw`	5,013.16	7,665.92	0.0	32,947.0	156.0	56.8
US	`load_mw`	75,757.21	11,819.46	52,940.0	120,343.0	73,564.0	100.0
US	`wind_mw`	10,955.94	6,041.83	0.0	26,132.0	10,183.5	100.0
US	`solar_mw`	2,875.42	3,842.56	0.0	14,315.0	110.0	94.6

3.6 Leakage-Safe Temporal Framing

Both DE and US training configs use a 24-hour forecast horizon and 168-hour lookback with time-aware cross-validation and an explicit 24-hour gap (`cross_validation.n_folds = 10`, `cross_validation.gap = 24`). This temporal framing reduces leakage risk and protects downstream decision-impact validity.

4 Methods

4.1 Forecasting Method

Model Families

GridPulse trains three forecasting families per target:

1. GBM (LightGBM), used as the primary operational model in the locked dashboard snapshots.
2. LSTM sequence model.
3. TCN sequence model.

Training Setup

From `configs/train_forecast.yaml` and `configs/train_forecast_eia930.yaml`:

- Forecast horizon: 24 hours.
- Lookback context: 168 hours.
- Time-aware CV configured (10 folds, 24-hour gap).
- Seeded, reproducible training controls.

Implementation notes important for reproducibility:

1. Both regions keep GBM enabled as baseline operational model with explicit LightGBM parameter blocks.
2. LSTM and TCN blocks are present in both configs, with region-specific parameterization.
3. Cross-validation behavior is explicit in config rather than implicit defaults.
4. Training outputs are persisted into region-scoped artifact/report paths.

Runtime training controls are explicit in `src/gridpulse/forecasting/train.py` and `scripts/train_dataset.py`. `-tune` force-enables Optuna tuning, `-no-tune` disables tuning even when enabled in YAML, `-ensemble` trains multi-seed GBM members, `-max-seeds` caps ensemble size, and `-n-trials`/`-top-pct` override trial-count and top-trial aggregation behavior for tuned runs. Saved bundles persist selected tuning metadata and `ensemble_models`/seed-member diagnostics so inference and report generation remain run-reproducible.

Metric Interpretation

Primary comparison metrics are RMSE, MAE, sMAPE, and R^2. MAPE is recorded but can be numerically unstable in low-denominator regimes (notably solar nighttime and near-zero wind segments), so interpretation prioritizes sMAPE and absolute error metrics. Interpretation policy:

1. RMSE and MAE are primary for magnitude-sensitive operational quality.
2. sMAPE supports relative comparability across differently scaled targets.
3. R^2 is retained as fit context, not a standalone decision metric.
4. MAPE is treated as supplementary in near-zero target regimes.

Forecast Inference Contract

API-level forecast inference (`services/api/routers/forecast.py`) follows a strict contract:

1. Load features from configured path.
2. Resolve model bundle by explicit or fallback target mapping.
3. Emit next-horizon forecast outputs with metadata.
4. Surface missing-model states explicitly (not silent fallback fabrication).

Resolution behavior is deterministic: explicit per-target model paths are checked first, then ordered fallback search (`lstm`, `tcn`, `gbm`) is applied under `artifacts/models`. If `ensemble_models` is present in a GBM bundle, inference averages member predictions; quantiles are emitted from explicit quantile models when present, otherwise from stored residual quantile offsets.

4.2 Uncertainty: Conformal and Adaptive Intervals

Conformal calibration at $\alpha = 0.10$ is implemented in the uncertainty module (`conformal.py`). For point forecast $\hat{y}$ and calibrated residual quantile q, interval bounds are:

$$\text{Lower} = \hat{y} - q, \qquad \text{Upper} = \hat{y} + q.$$

The implementation supports horizon-wise quantiles, rolling updates, and adaptive scaling behavior for online coverage control. It combines two mechanisms: `ConformalInterval` for calibrated interval construction and `AdaptiveConformal` for bounded online alpha updates under miss events. Default code-level settings include `alpha = 0.10`, `rolling = true`, and `rolling_window = 720`.

4.3 Deterministic Dispatch Optimization

The deterministic optimizer in `src/gridpulse/optimizer/lp_dispatch.py` solves a mixed-integer linear formulation with:

- battery charge/discharge decisions,
- grid import,
- SOC dynamics and constraints,
- cost and carbon objective terms,
- degradation and peak penalties.

Representative objective form:

$$\min \sum_t \left(c_t^{\text{energy}} G_t + c_t^{\text{carbon}} G_t + c^{\text{deg}} (P_t^{\text{ch}} + P_t^{\text{dis}}) \right).$$

Key implementation behaviors:

1. Piecewise battery efficiency regimes (`efficiency_regime_a`, `efficiency_regime_b`) with SOC split gating.
2. Optional interval-aware risk binding via `risk.mode = worst_case_interval`.
3. High unmet-load and curtailment penalties to avoid unsafe economic shortcuts.

Locked default config highlights in `configs/optimization.yaml`: `objective.cost_weight = 1.0`, `objective.carbon_weight = 1.2`, battery capacity 20,000 MWh, battery max power 5,000 MW, grid import cap 100,000 MW.

4.4 Robust Dispatch Optimization

The robust optimizer in `src/gridpulse/optimizer/robust_dispatch.py` solves a two-scenario min-max linear program over lower/upper load bounds. With epigraph variable z and scenario costs C_s:

$$\min \lambda z + (1 - \lambda)\bar{C} + c^{\text{deg}} \text{throughput}, \qquad z \geq C_s \ \forall s.$$

The solver returns feasibility flags, scenario costs, and dispatch trajectories for analysis. Additional behavior:

1. Charge/discharge decisions are shared across scenarios; grid import and SOC remain scenario-indexed.
2. Non-optimal termination returns explicit infeasible-safe payloads rather than partial unsafe commands.
3. Solver availability and status are explicit in outputs used for publication metrics.

4.5 DC3S: Drift-Calibrated Conformal Safety Shield

DC3S is implemented in `src/gridpulse/dc3s/` and exposed through `POST/dc3s/step` and `GET/dc3s/audit/\{command\_id\}` in `services/api/routers/dc3s.py`. It composes telemetry quality weighting, drift detection, uncertainty inflation, safe action repair, and certificate-grade audit persistence.

The implemented interval inflation law is linear and bounded:

$$\text{infl} = \text{clip}\left(1 + k_q(1 - w_t) + k_d \mathbf{1}[\text{drift}],\ 1,\ \text{infl}_{\max}\right)$$

$$\text{Lower} = \hat{y} - q \cdot \text{infl}, \qquad \text{Upper} = \hat{y} + q \cdot \text{infl}$$

$$w_t \leftarrow \max(w_t, \min_w)$$

The reliability module (`quality.py`) computes w_t from missingness, delay, out-of-order behavior, and spike penalties. Drift logic (`drift.py`) uses Page-Hinkley updates on residual magnitude $r_t = |y_t - \hat{y}_t|$ with configured warmup and cooldown controls.

Safety shielding (`shield.py`) supports two modes:

1. `projection`: deterministic clipping against SOC, power, and ramp constraints.
2. `robust_resolve`: attempt robust optimization under interval bounds, then project as final guard.

Certification (`certificate.py`) computes `model_hash` and `config_hash`, supports optional `prev_hash` chaining, and stores signed payload structure to DuckDB. Runtime online state (`state.py`) persists per `(zone, device, target)` keys including drift state, last inflation, latest telemetry/action context, and audit linkage pointers.

4.6 Monitoring and Retraining Controls

Monitoring logic combines data-drift and model-drift checks:

- KS-based feature distribution drift in `src/gridpulse/monitoring/data_drift.py`.
- Model degradation thresholds with retraining decisions in the monitoring retraining module.
- API monitoring endpoints in `services/api/routers/monitor.py`.

Configured default thresholds in the locked snapshot:

1. `data_drift.p_value_threshold = 0.01`
2. `model_drift.metric = mape`
3. `model_drift.degradation_threshold = 0.15`
4. `retraining.cadence_days = 30`
5. `retraining.min_new_data_days = 14`

4.7 Safety and Operational Controls

Safety mechanisms include:

1. BMS validation for dispatch commands.
2. Watchdog heartbeats and island-mode protection.
3. API key scope checks for read/write control routes.
4. Health and readiness endpoints for deployment gating.

The `/control/dispatch` route is protected by three gates: write-scope authorization, watchdog islanding check, and BMS constraint validation on SOC/power commands.

4.8 Reproducibility and Governance Controls

Publication claims are locked via:

- `paper/metrics_manifest.json`
- `paper/claim_matrix.csv`
- `scripts/validate_paper_claims.py`

These are release gates, not passive documentation: a manuscript change is incomplete unless validator checks pass and claim provenance remains traceable.

4.9 Method Assumptions and Boundary Conditions

This thesis assumes hourly timestep consistency, proxy suitability of price/carbon channels for comparative decision analysis in locked windows, interval-based uncertainty representation, and configured battery/grid constraints sufficient for physically plausible dispatch plans. These are engineering assumptions, not universal truths.

4.10 Baselines and Comparator Policies

This manuscript uses three explicit baseline policies for decision comparisons:

1. **B1 (Deterministic LP, no uncertainty intervals):** implemented in `src/gridpulse/optimizer/lp_dispatch.py` via `optimize_dispatch(...)` without `load_interval` / `renewables_interval`. It uses battery + grid constraints and is the optimization policy reported as GridPulse in canonical impact artifacts.
2. **B2 (Grid-only, no battery):** implemented in `src/gridpulse/optimizer/baselines.py` as `grid_only_dispatch(...)`. It sets `battery_charge_mw=0` and `battery_discharge_mw=0` and serves net deficit from grid import. This is the denominator baseline for canonical DE/US impact percentages.
3. **B3 (Naive battery heuristic):** implemented in `src/gridpulse/optimizer/baselines.py` as `naive_battery_dispatch(...)` with a fixed rule that charges at 00–05 and discharges at 17–21.

Runtime disambiguation: while API `/optimize` may default to robust mode in service context, publication impact percentages in this thesis are produced from offline deterministic `optimize_dispatch(...)` (B1) versus grid-only baseline (B2).

5 Experimental Protocol

5.1 Evidence-Lock Policy

All publication-facing values in this thesis must map to canonical entries in `paper/metrics_manifest.json`. The markdown master file is `paper/PAPER_DRAFT.md`.

5.2 Canonical Source and Run Selection

5.3 Rounding and Formatting Rules

1. Percent values: `round(value*100,2)`.
2. Monetary/mass/power values: two-decimal display where reported.
3. Large stochastic values: thousands separators plus two decimals.

Table 5: Canonical Source-of-Truth Mapping

Claim Domain	Source Artifact	Selection Rule
DE decision impact	`reports/impact_summary.csv`	latest dataset-scoped record
US decision impact	`reports/eia930/impact_summary.csv`	latest dataset-scoped record
DE stochastic metrics	`reports/research_metrics_de.csv`	`run_id=20260217_165756`, `row_type=run_summary`
US stochastic metrics	`reports/research_metrics_us.csv`	`run_id=20260217_182305`, `row_type=run_summary`
Dataset profile claims	`data/dashboard/*_stats.json` + manifest	dashboard profile lock `2026-02-17T11:15:38.623283`

5.4 Cross-Region Comparability Rules

1. Compare methods and process governance across regions.
2. Do not enforce equal effect-size expectations across regions.
3. Keep all stochastic claims tied to region-specific run IDs.

5.5 Consistency Gates

The manuscript is release-ready only when:

1. `pythonscripts/validate_paper_claims.py` passes.
2. Required run IDs and canonical values are present in markdown and LaTeX.
3. No banned legacy patterns or placeholders appear.
4. Publication claims map to `paper/claim_matrix.csv`.

5.6 Reconciliation Procedure for Conflicts

When a conflicting claim is found:

1. Locate claim text and claim ID in `paper/claim_matrix.csv`.
2. Identify canonical source and extraction rule in `paper/metrics_manifest.json`.
3. Replace manuscript value using the locked rounding policy.
4. Re-run `pythonscripts/validate_paper_claims.py`.
5. Mirror correction in markdown, LaTeX, and DOCX sync outputs.

This blocks partial fixes where one format is updated and another remains stale.

6 Results

6.1 Forecast Quality Across Model Families

Source: `data/dashboard/de_metrics.json`, `data/dashboard/us_metrics.json`.

Table 6: Forecast Metrics by Model and Target (DE)

Target	Model	RMSE	MAE	sMAPE (%)	R^2
`load_mw`	GBM	267.51	167.49	0.35	0.9991
`load_mw`	LSTM	3,474.78	2,722.50	5.38	0.8498
`load_mw`	TCN	2,668.05	2,031.12	3.94	0.9114
`wind_mw`	GBM	183.93	118.19	2.29	0.9993
`wind_mw`	LSTM	6,735.07	5,511.23	60.85	0.1545
`wind_mw`	TCN	9,196.16	7,167.41	76.67	-0.5763
`solar_mw`	GBM	251.44	121.22	69.27	0.9992
`solar_mw`	LSTM	4,079.38	2,835.74	109.44	0.8007
`solar_mw`	TCN	2,702.34	1,583.05	93.76	0.9125

Germany (DE)

United States (US)

Table 7: Forecast Metrics by Model and Target (US)

Target	Model	RMSE	MAE	sMAPE (%)	R^2
`load_mw`	GBM	162.89	123.23	0.17	0.9996
`load_mw`	LSTM	4,767.14	3,835.32	5.24	0.6014
`load_mw`	TCN	3,850.49	2,877.31	3.82	0.7399
`wind_mw`	GBM	269.23	144.66	1.75	0.9982
`wind_mw`	LSTM	6,301.91	5,234.31	42.59	0.0627
`wind_mw`	TCN	7,187.54	5,930.74	47.65	-0.2192
`solar_mw`	GBM	208.92	74.93	45.45	0.9962
`solar_mw`	LSTM	1,781.97	1,055.34	136.92	0.7135
`solar_mw`	TCN	1,743.66	965.08	131.50	0.7257

GBM is the strongest model family in the locked operational snapshots across both regions and all targets. Thesis-level interpretation:

1. GBM leads across load, wind, and solar in both regions for the locked artifacts.
2. Deep models remain informative but are not operational leaders in this snapshot.
3. Forecast rankings should be read together with downstream optimization outcomes.

6.2 GBM Residual and Coverage Diagnostics (90% Intervals)

Table 8: Residual Quantiles and Coverage by Region/Target

Region	Target	Residual q10	Residual q50	Residual q90	PICP (%)
DE	`load_mw`	-321.56	-11.61	254.52	95.17
DE	`wind_mw`	-293.49	-42.68	221.95	88.15
DE	`solar_mw`	-165.27	0.40	336.66	87.00
US	`load_mw`	-240.08	1.00	223.77	87.39
US	`wind_mw`	-146.14	-3.96	122.64	80.50
US	`solar_mw`	-105.40	-0.14	186.94	90.47

Coverage behavior is near nominal for selected targets and lower for others, indicating region-target calibration opportunities. Coverage pattern notes:

1. DE load coverage is conservative relative to 90% nominal.

2. US wind coverage is materially below nominal and indicates under-coverage risk.

3. Solar coverage differs across regions, reflecting profile and feature differences.

6.3 Baselines and Comparator Mapping (6.2A)

Table 9: Baseline Mapping by Results Table

Result section/table	Numerator policy	Denominator/comparator policy	Source artifact
Decision Impact (Canonical Percent Outcomes)	B1 deterministic LP (no uncertainty intervals)	B2 grid-only (no battery)	`reports/impact_summary.csv`; `reports/eia930/impact_summary.csv`
Decision Impact (Absolute Values)	B1 deterministic LP (no uncertainty intervals)	B2 grid-only (no battery)	`reports/impact_summary.csv`; `reports/eia930/impact_summary.csv`
Stochastic Value Metrics (Canonical Run IDs)	robust vs deterministic stochastic formulations	deterministic/robust run-family comparator (not B2 denominator)	`reports/research_metrics_de.csv`; `reports/research_metrics_us.csv`

6.4 Baseline Absolute Policy Outcomes (DE/US) (6.2B)

Source files: `reports/impact_comparison.json` (DE), `reports/eia930/impact_comparison.json` (US). The table uses existing serialized artifacts only (`baseline` = B2, `naive` = B3, `optimized_forecast` = B1) with two-decimal display rounding.

Table 10: Baseline Absolute Outcomes by Region and Policy

Region	Policy ID	Policy name	Cost (USD)	Carbon (kg)	Peak (MW)
DE	B2	Grid-only (no battery)	154,773,462.72	1,461,641,243.28	52,165.00
DE	B3	Naive battery heuristic	154,200,945.64	1,454,657,899.03	52,165.00
DE	B1	Deterministic LP (no uncertainty intervals)	143,776,429.88	1,457,205,025.31	48,969.37
US	B2	Grid-only (no battery)	461,364,890.00	3,907,803,618.32	74,741.00
US	B3	Naive battery heuristic	462,821,035.03	3,911,194,390.28	70,929.00
US	B1	Deterministic LP (no uncertainty intervals)	460,842,677.76	3,902,642,369.72	74,741.00

6.5 Decision Impact (Canonical Percent Outcomes)

Table 11: Canonical Decision Impact Percentages

Region	Cost Savings	Carbon Reduction	Peak Shaving
DE	7.11%	0.30%	6.13%
US	0.11%	0.13%	0.00%

Source files: `reports/impact_summary.csv`, `reports/eia930/impact_summary.csv`. All reported DE/US impact percentages are computed as B1 vs B2. Reading guidance:

1. Percent values are canonical publication claims.

2. Absolute outcomes provide scale interpretation.

3. DE and US should not be treated as directly interchangeable market contexts.

6.6 Decision Impact (Absolute Values)

Absolute outcomes show that modest percentages in large systems can still correspond to substantial absolute deltas.

Table 12: Absolute Cost and Carbon Outcomes

Region	Baseline Cost (USD)	GridPulse Cost (USD)	Cost Delta (USD)	Baseline Carbon (kg)	GridPulse Carbon (kg)	Carbon Delta (kg)
DE	154,773,462.72	143,776,429.88	10,997,032.84	1,461,641,243.28	1,457,205,025.31	4,436,217.97
US	461,364,890.00	460,842,677.76	522,212.24	3,907,803,618.32	3,902,642,369.72	5,161,248.60

Table 13: Absolute Peak Outcomes

Region	Baseline Peak (MW)	GridPulse Peak (MW)	Peak Delta (MW)
DE	52,165.00	48,969.37	3,195.63
US	74,741.00	74,741.00	0.00

6.7 Stochastic Value Metrics (Canonical Run IDs)

US run-summary operational controls in the same canonical row include grid cap 69,844.27 MW, reserve SOC 2,000.00 MWh, terminal SOC target 4,450.43 MWh, load stress additive 1,257.02 MW, and stress interval multiplier 1.90. Interpretation:

1. Both canonical runs are feasible and explicitly tied to run IDs and timestamps.
2. DE run has modest EVPI magnitudes with positive VSS.
3. US run has large EVPI magnitudes with positive VSS, indicating strong uncertainty sensitivity.
4. These statements are run-scoped and should always include run IDs in text.

6.8 Robustness Perturbation Summary

Source: `reports/publication/tables/table6_robustness.csv`.

All listed perturbation points remain feasible in this summary table. Regret sign caution:

1. Negative regret values can occur depending on comparator definition and realization.
2. Regret interpretation must use consistent source-table definitions.

6.9 Cross-Region Interpretation

1. Both DE and US show positive VSS under canonical locked runs.
2. Magnitude differences are substantial, indicating sensitivity to regime and scenario design.
3. DE shows stronger direct percent impact in this snapshot.
4. US shows smaller direct percent impact but large stochastic value magnitudes.

This supports a core thesis claim: forecast-to-dispatch systems should be evaluated as region-specific decision systems, not only by aggregate forecast-score transferability.

Table 14: Canonical Stochastic Metrics by Locked Run

Region	Run ID	Timestamp (UTC)	$\text{EVPI}_{\text{robust}}$	$\text{EVPI}_{\text{deterministic}}$	VSS	Feasible	Solver
DE	`20260217_165756`	2026-02-17T16:57:56.736552	2.32	-30.40	2,708.61	True	summary
US	`20260217_182305`	2026-02-17T18:23:06.287827	10,279,851.74	24,915,503.93	297,092.71	True	summary

Table 15: Additional Run-Summary Context

Region	Mean Dynamic Width	Mean Base Width	Mean Stressed Width	FACI Scale Lower	FACI Scale Upper
DE	653.57	657.11	not reported in DE schema	0.0648	1.0000
US	649.20	415.60	1,233.49	0.1035	1.5965

Table 16: Perturbation Robustness Summary

Dataset	Perturbation (%)	Infeasible (%)	Mean Regret
DE	0.0	0.0	0.00
DE	5.0	0.0	-1,509.47
DE	10.0	0.0	-68,063.96
DE	20.0	0.0	-183,810.02
DE	30.0	0.0	-142,933.97
US	0.0	0.0	0.00
US	5.0	0.0	25,431.64
US	10.0	0.0	-241,341.60
US	20.0	0.0	-140,118.36
US	30.0	0.0	506,266.38

6.10 Copy-Ready Core Claims

Use these exact strings to avoid drift:

1. DE cost savings: 7.11%; DE carbon reduction: 0.30%; DE peak shaving: 6.13%.
2. US cost savings: 0.11%; US carbon reduction: 0.13%; US peak shaving: 0.00%.
3. DE stochastic run: 20260217_165756 ($\text{EVPI}_{\text{robust}}$ 2.32; $\text{EVPI}_{\text{deterministic}}$ -30.40; VSS 2,708.61).
4. US stochastic run: 20260217_182305 ($\text{EVPI}_{\text{robust}}$ 10,279,851.74; $\text{EVPI}_{\text{deterministic}}$ 24,915,503.93; VSS 297,092.71).
5. Dataset-scoped latest policy is used; this manuscript is not a single common-run freeze.

6.11 Negative and Neutral Findings (Do Not Omit)

1. US peak shaving in locked impact sources is 0.00%.
2. Several interval coverage values fall below 90% nominal target.
3. Deep models are not top-ranked in locked dashboard snapshots.
4. Cross-region impact magnitude differs significantly despite shared architecture.

7 Metrics Source of Truth and Run IDs

7.1 Locked Sources

7.2 Update Rule for Any Future Metric Change

A manuscript metric can be changed only if all conditions hold:

1. Value is updated in `paper/metrics_manifest.json` with source path and run ID.
2. Claim row exists or is updated in `paper/claim_matrix.csv`.
3. `pythonscripts/validate_paper_claims.py` passes after edit.

Table 17: Locked Source-of-Truth Artifacts

Claim Group	Canonical Source
DE impact metrics	`reports/impact_summary.csv`
US impact metrics	`reports/eia930/impact_summary.csv`
DE stochastic metrics	`reports/research_metrics_de.csv`, `run_id = 20260217_165756`
US stochastic metrics	`reports/research_metrics_us.csv`, `run_id = 20260217_182305`
Dataset profile claims	`data/dashboard/de_stats.json`, `data/dashboard/us_stats.json`, `data/dashboard/manifest.json`

4. Markdown and LaTeX manuscripts retain matching core metric statements.

This applies to percentages, absolute values, run IDs, timestamps, and any sentence embedding quantitative evidence.

7.3 Non-Policy Sources

Legacy report exports and older markdown writeups are treated as secondary references unless explicitly reconciled to this lock.

7.4 Metrics Change-Control Checklist

Before approving a numeric edit:

1. Confirm source artifact path exists and is versioned.
2. Confirm row and field selection rule is unambiguous.
3. Regenerate display values from raw values using manifest rounding rules.
4. Verify metric synchronization in abstract, results, and conclusion.
5. Re-run validator and archive pass output in revision notes.

8 Claim Traceability and Editorial Governance

8.1 Claim Status Definitions

- **Verified**: directly traceable to locked source artifacts.
- **Conflicting**: contradicts canonical policy or locked values.
- **Unsupported**: no in-repo evidence found.
- **Needs Citation**: plausible but requires external source citation.

Claim statuses are operational controls; status changes require evidence/citation change, not wording-only edits.

8.2 Traceability Fields per Claim

Each claim entry in `paper/claim_matrix.csv` should include:

1. `claim_id`
2. `status`
3. `category`
4. `claim_text`
5. `source_file`
6. `source_locator`
7. `run_id` (when applicable)
8. `timestamp_utc` (when applicable)
9. `rounding_rule`

8.3 Publication Rule

Publication-facing sections should contain only:

1. Verified in-repo claims.
2. External claims that are explicitly cited.

8.4 Editing Workflow

1. Edit `paper/PAPER_DRAFT.md` as the master.
2. Run the claim validator.
3. Synchronize `paper/paper.tex`.
4. Synchronize DOCX last and record any environment/tooling limitations.

8.5 Reviewer and Advisor Workflow

1. Reviewer flags text spans or claim IDs.
2. Author resolves each item against `paper/claim_matrix.csv` and manifest sources.
3. Any unresolved numeric conflict blocks release.
4. Validator pass output is attached to the revision package.

9 Discussion

9.1 Strongly Supported Findings

1. DE and US impact percentages are directly reproducible from canonical impact CSVs.
2. DE and US stochastic values are directly reproducible from canonical run-summary rows.
3. GBM is the top forecast performer in locked dashboard snapshots.
4. Governance controls materially reduce manuscript drift risk.

The strongest thesis claim is the combined result of measurable decision outcomes and explicit provenance enforcement.

9.2 Why Cross-Region Outcomes Differ

Likely contributors include differences in data window, operating context, feature-space composition, and stress/scenario settings. This thesis reports region-scoped outcomes and does not merge them into a single universal effect claim. Deployment implication: a strategy with high percentage gains in one region can remain valuable in another while producing smaller direct percentages.

9.3 Non-Overclaim Boundary

The thesis does not claim universal superiority across all grids. All quantitative statements are tied to the locked DE/US evidence scope.

9.4 Engineering Significance

The primary engineering contribution is integrated, auditable decision evidence: forecasting, uncertainty, optimization, and governance are connected in one reproducible workflow.

9.5 Practical Operator Interpretation

Operationally, GridPulse is a guarded decision-support stack:

1. Healthy forecast and interval artifacts trigger optimization-driven dispatch plans.
2. Infeasible or degraded states trigger explicit safe fallbacks.
3. Watchdog and BMS controls enforce safety before dispatch acceptance.

Operationally, the stack is ready for software deployment pathways and controlled integration tests, but hardware field validation remains pending in the current evidence lock.

10 Threats to Validity

10.1 Internal Validity

Risk: metric drift from mixed artifacts. Mitigation: manifest lock, claim matrix, and automated validator checks.

10.2 External Validity

Risk: limited transferability beyond locked DE/US windows. Mitigation: explicit scope boundaries and future multi-region validation.

10.3 Construct Validity

Risk: proxy price/carbon signals may not represent all market settlement details. Mitigation: transparent assumptions and conservative interpretation.

10.4 Conclusion Validity

Risk: stochastic value magnitudes are sensitive to run/scenario settings. Mitigation: run-ID pinning and explicit source traceability.

10.5 Residual Risk Prioritization

1. Preserve strict metric synchronization across markdown, LaTeX, and DOCX.
2. Improve calibration on under-covered targets.
3. Expand external validity with additional regions/time windows.
4. Close citation gaps for non-repository claims.

11 Operational Safety and Failure Modes

11.1 Failure-Mode Table

Table 18: Operational Failure Modes and Controls

Failure Mode	Trigger	Detection	Mitigation	Fallback
Missing/stale data	upstream delay or schema break	readiness checks and file checks	block unsafe requests until fresh inputs exist	health-warning state
Forecast drift	distributional shift	KS/model drift monitors	retraining decision workflow	conservative dispatch mode
Optimization infeasible	stress or malformed intervals	solver status and feasibility flags	safe infeasible response payload	grid-only baseline dispatch
Unsafe dispatch command	SOC/power violation	BMS validation	reject command with error detail	maintain current safe state
Control degradation	heartbeat timeout	watchdog island detection	lock remote control route	manual/local control
API auth misuse	invalid scope/key	security middleware	deny operation	read-only status routes

11.2 Incident Response Sequence

1. Detect (health, monitor, solver status).
2. Contain (block unsafe dispatch, switch fallback mode).
3. Diagnose (trace artifact/run lineage).
4. Recover (retrain, reconfigure, or rollback).
5. Record (update governance artifacts and incident notes).

11.3 Governance Safeguards

1. Locked source-of-truth artifacts.
2. Validator-enforced banned legacy values.
3. Claim-level status and provenance tracking.

11.4 Config-Level Safety Snapshot

Selected defaults in locked configs:

1. KS drift threshold: `p_value_threshold = 0.01`.
2. Model drift threshold: `degradation_threshold = 0.15`.
3. Retraining cadence: 30 days with minimum new-data rule.
4. Deterministic objective weights: cost 1.0, carbon 1.2.
5. High unmet-load and curtailment penalties for safe optimization behavior.

12 Limitations

12.1 Dataset and Coverage Limits

Evidence is limited to the locked DE/US windows in this manuscript version. These limits are intentional for traceability and should only be expanded with corresponding manifest and claim-matrix updates.

12.2 Modeling and Optimization Limits

Stochastic value magnitudes are sensitive to scenario/stress choices and proxy assumptions. Robust objective weighting and interval-construction choices can shift tradeoff behavior, so results are evidence for this configured system rather than universal constants.

12.3 Documentation Synchronization Limits

Current shell limitations constrain programmatic DOCX synchronization:

- system `python3` lacks `python-docx`,
- local `.venv` launcher is currently unusable.

12.4 Citation Limits

Broader operational and societal claims require external source citation before publication. Claims without internal artifact evidence should remain explicitly marked as requiring citation.

12.5 IoT Validation Boundary (Deployment-Readiness, Non-Field Evidence)

IoT-related validation reported in this manuscript is software-in-the-loop/API/streaming-smoke evidence, not completed hardware field commissioning evidence.

1. Streaming checks rely on replayed telemetry and broker-consumer pipeline execution.
2. DC3S live checks are API-level evaluations driven by simulated/replayed telemetry inputs.
3. No physical inverter/BMS edge commissioning logs or field-trial artifacts are claimed in this revision.

Therefore, deployment-readiness claims are limited to software stack correctness and observability, not hardware-operational certification.

13 Future Work

13.1 Near-Term

1. Unify report generation under one profile-lock workflow.
2. Add stricter claim-language linting for unsupported statements.
3. Improve interval calibration for under-covered targets.

Success condition: no unresolved claim conflicts and synchronized core metrics across markdown/LaTeX outputs.

13.2 Mid-Term

1. Extend stochastic scenario sensitivity studies.
2. Add more regions and tariff/carbon regimes.
3. Enhance model registry lineage and release metadata.

Success condition: cross-region comparisons with fully serialized scenario assumptions and reproducible run-level evidence.

13.3 Long-Term

1. Move to signed evidence bundles for release claims.
2. Integrate richer market and operational constraints.
3. Automate publication synchronization with deterministic release gates.

Success condition: signed provenance from data extraction to publication-ready manuscript artifacts.

14 Conclusion

GridPulse delivers a decision-grade ML system connecting forecasting, uncertainty, dispatch optimization, and governance. Under the dataset-scoped latest lock (February 17, 2026 artifacts), DE impact is 7.11% cost savings, 0.30% carbon reduction, and 6.13% peak shaving; US impact is 0.11% cost savings, 0.13% carbon reduction, and 0.00% peak shaving. Canonical stochastic values are tied to DE run `20260217_165756` and US run `20260217_182305`, with positive VSS in both regions (2,708.61 and 297,092.71). The thesis contribution is therefore both technical and operational: measurable decision outcomes plus reproducible claim governance. The practical message is that trustworthy decision systems require both technical performance and claim governance; GridPulse integrates forecast quality, uncertainty calibration, optimization feasibility, and publication traceability as one coherent system. Current IoT validation evidence in this manuscript is non-field and should be interpreted as software stack readiness rather than completed hardware-operational certification.

References

Internal Artifact References

- paper/metrics_manifest.json
- paper/claim_matrix.csv
- paper/accuracy_audit.md
- paper/rewrite_pack.md
- paper/sync_rules.md
- scripts/validate_paper_claims.py
- frontend/src/app/api/chat/tool-executors.ts
- services/api/routers/monitor.py
- src/gridpulse/streaming/run_consumer.py
- src/gridpulse/streaming/consumer.py
- src/gridpulse/streaming/worker.py
- configs/dc3s.yaml
- src/gridpulse/dc3s/
- services/api/routers/dc3s.py
- data/dashboard/manifest.json
- data/dashboard/de_stats.json
- data/dashboard/us_stats.json
- data/dashboard/de_metrics.json
- data/dashboard/us_metrics.json
- reports/impact_summary.csv
- reports/eia930/impact_summary.csv
- reports/impact_comparison.json
- reports/eia930/impact_comparison.json
- reports/research_metrics_de.csv
- reports/research_metrics_us.csv
- reports/publication/tables/table6_robustness.csv
- docs/ARCHITECTURE.md
- docs/TRAINING_PIPELINE.md
- docs/EVALUATION.md

External Method Literature (Maintain in Publication Export)

Conformal prediction, adaptive conformal inference, robust optimization, and time-series forecasting method references should be retained in the publication export.

External Factual Claims Requiring Citation

Any non-repository factual claim (policy forecasts, staffing claims, environmental lifecycle values) requires explicit external citation before publication.

A Replication Checklist

1. Validate manuscript claims:

   ```
   python scripts/validate_paper_claims.py
   ```

2. Compile LaTeX manuscript:

   ```
   cd paper
   pdflatex paper.tex
   ```

3. Confirm canonical strings exist in markdown and LaTeX:

   - `7.11%`
   - `0.11%`
   - `20260217_165756`
   - `20260217_182305`
   - `2,708.61`
   - `297,092.71`

4. Confirm no banned legacy percentages or placeholders.

5. Confirm publication numeric claims map to `paper/claim_matrix.csv`.

B Artifact Inventory

C Section-by-Section Writing Bank (Copy-Ready)

C.1 Introduction

Required facts:

1. Forecast-only systems are insufficient for operations.

2. GridPulse decision-loop framing.

3. Evidence-lock policy and date.

Optional additions: deployment context and operational motivation.

Table 19: Core Governance and Evidence Artifacts

Artifact	Purpose
`paper/PAPER_DRAFT.md`	Master manuscript source
`paper/paper.tex`	LaTeX manuscript export
`paper/metrics_manifest.json`	Canonical metric lock and validation rules
`paper/claim_matrix.csv`	Claim-level provenance and status
`paper/accuracy_audit.md`	Accuracy/conflict audit log
`paper/rewrite_pack.md`	Detailed section guidance bank
`paper/sync_rules.md`	Markdown-to-LaTeX-to-DOCX synchronization contract
`scripts/validate_paper_claims.py`	Non-mutating cross-manuscript consistency checker
`data/dashboard/*`	Dataset profiles and dashboard metrics
`reports/impact_summary.csv`	Canonical DE impact outcomes
`reports/eia930/impact_summary.csv`	Canonical US impact outcomes
`reports/research_metrics_de.csv`	DE stochastic run outputs
`reports/research_metrics_us.csv`	US stochastic run outputs
`reports/publication/tables/table6_robustness.csv`	Perturbation robustness summary

C.2 Data Assets and Scope

Required facts:

1. DE and US rows, date ranges, and feature counts.
2. Distinction between raw columns and engineered features.
3. Legacy-scope exclusion rule.

Required tables: dataset profile and feature family composition.

C.3 Methods

Required facts:

1. Forecast model families and horizon setup.
2. Conformal and adaptive interval logic.
3. Deterministic and robust optimization objectives/constraints.
4. Monitoring and safety implementation controls.

C.4 Results

Required facts:

1. Forecast metrics by model and target.
2. Coverage and residual diagnostics.
3. Canonical impact percentages and absolute outcomes.
4. Canonical stochastic values with run IDs.
5. Robustness perturbation summary.

C.5 Governance Sections

Required facts:

1. Source-of-truth artifact map.
2. Claim status model and editorial rule.
3. Validation gates and synchronization workflow.

C.6 Safety, Validity, and Limitations

Required facts:

1. Operational failure modes and mitigations.
2. Internal/external/construct/conclusion validity boundaries.
3. Current environment limitations for DOCX sync.

D Claims Requiring Citation Before Publication

The following claim classes require external source citation or removal in publication export:

1. Global market-size or policy forecasts.
2. Named-operator staffing claims.
3. Environmental footprint values without reproducible in-repo method.
4. Universal superiority statements beyond locked DE/US evidence.

E Synchronization and Release Notes

1. `paper/PAPER_DRAFT.md` remains the authority manuscript.
2. `paper/paper.tex` must preserve identical title, abstract core metrics, and conclusion core metrics.
3. DOCX synchronization is performed last and must record tooling limits when automation is unavailable.
4. Final release gate requires validator pass and no unresolved publication-facing claim conflicts.

F Full-Detail Authoring Blocks (Use/Trim as Needed)

F.1 Long-Form Intro Block

GridPulse is designed around a decision-first interpretation of machine learning in energy systems. Rather than ending at point prediction, the system carries uncertainty information into optimization, enforces dispatch feasibility under battery and grid constraints, and evaluates outcomes against explicit baselines. This architecture creates a measurable bridge between model behavior and operational value.

F.2 Long-Form Method Block

The forecasting layer trains LightGBM, LSTM, and TCN families under time-aware configurations with a 24-hour horizon and 168-hour lookback. Uncertainty is represented using conformal intervals with adaptive behavior. Deterministic optimization uses a mixed-integer structure with cost, carbon, degradation, curtailment, unmet-load, and peak terms. Robust optimization uses a two-scenario epigraph formulation over lower and upper load bounds, and safety checks are enforced before dispatch control actions are accepted.

F.3 Long-Form Results Block

Under the locked artifact policy, DE impact is 7.11% cost savings, 0.30% carbon reduction, and 6.13% peak shaving. US impact is 0.11% cost savings, 0.13% carbon reduction, and 0.00% peak shaving. Canonical stochastic evidence is tied to DE run `20260217_165756` and US run `20260217_182305`; both runs are feasible in run-summary outputs. Positive VSS appears in both regions with substantially different magnitudes.

F.4 Long-Form Governance Block

The manuscript uses a source-of-truth contract where publication-facing numeric claims must map to `paper/metrics_manifest.json` and corresponding entries in `paper/claim_matrix.csv`. Validation is automated via `scripts/validate_paper_claims.py`, which enforces required canonical strings and blocks known legacy contradictions.

F.5 Long-Form Limitation Block

Evidence is intentionally restricted to locked DE and US windows from dashboard profile artifacts. This improves traceability but limits direct external generalization. Optimization outcomes also depend on proxy price/carbon assumptions and scenario/interval settings that may vary across market designs.

F.6 Long-Form Future Work Block

Future work should unify report-generation pathways, improve uncertainty calibration where coverage is below nominal, expand region and tariff coverage, and strengthen release automation with signed evidence bundles.

F.7 Keep/Trim Guidance for Final Submission

If page limits require reduction:

1. Keep Sections 6, 7, 8, and 14 unchanged.
2. Keep all canonical run IDs and percentages unchanged.
3. Trim descriptive implementation prose before trimming governance controls.
4. Preserve limitations and citation-boundary statements.

G Project-to-Paper Coverage Map

This appendix maps implemented project surfaces to manuscript coverage so review can distinguish fully documented areas from areas that are currently summarized.

Coverage status definitions:

1. **Full**: implementation paths and operational behavior are explicitly described in the paper.
2. **Partial**: core behavior is described, but implementation/runtime details are summarized.

Table 20: Project-to-Paper Coverage Matrix

Project Surface	Primary Implementation Paths	Current Paper Coverage	Status	Notes
Forecast training orchestration, splits, tuning	`src/gridpulse/forecasting/train.py`, `configs/train_forecast.yaml`, `configs/train_forecast_eia930.yaml`, `scripts/train_dataset.py`	Sections 3.7, 4.1	Full	Runtime control flags (`-tune`, `-no-tune`, `-ensemble`, `-max-seeds`, `-n-trials`, `-top-pct`) plus tuning/ensemble serialization behavior are explicitly documented.
Forecast inference and bundle resolution	`services/api/routers/forecast.py`, `src/gridpulse/forecasting/predict.py`	Sections 2.3, 4.1.4	Full	Explicit target-path resolution, ordered fallback search, ensemble averaging, quantile serving order, and missing-target signaling are documented.
Streaming ingestion and persistence	`src/gridpulse/streaming/run_consumer.py`, `src/gridpulse/streaming/consumer.py`, `src/gridpulse/streaming/worker.py`, `configs/streaming.yaml`	Sections 2.7, 12.5	Full	CLI path, config load path, and DuckDB table contract are explicitly documented.
DC3S uncertainty shield and audit chain	`src/gridpulse/dc3s/`, `services/api/routers/dc3s.py`, `configs/dc3s.yaml`	Sections 2.3, 2.7, 4.5, 12.5	Full	Endpoint contracts, implemented inflation law, shield modes, hash chain, and state/audit behavior are documented.
Monitoring and retraining	`services/api/routers/monitor.py`, `src/gridpulse/monitoring/`, `configs/monitoring.yaml`	Section 4.6	Full	Drift thresholds, cadence, and decision logic are explicitly documented.
Safety and control-plane enforcement	`src/gridpulse/safety/`, `services/api/main.py` (`/control/dispatch`)	Sections 4.7, 11	Full	Authorization, watchdog, and BMS gate sequence is documented.
Frontend operator workflow and live DC3S UX	`frontend/src/app/(dashboard)/page.tsx`, `frontend/src/components/dashboard/DC3SLiveCard.tsx`, `frontend/src/lib/api/dc3s-client.ts`, `frontend/src/app/api/dc3s/audit/[commandId]/route.ts`	Sections 2.2, 2.7	Full	Dashboard polling default, selectable auto-refresh cadence, manual refresh, command-linked audit quick link, and proxy/error behavior are explicitly documented.
Claim governance and release gates	`paper/metrics_manifest.json`, `paper/claim_matrix.csv`, `scripts/validate_paper_claims.py`	Sections 5, 7, 8	Full	Source lock, reconciliation, and publication gating are explicitly documented.

G.1 Full-Coverage Update (Completed)

This revision closes the previously partial coverage areas:

1. Section 4.1 now documents runtime training controls and their reproducibility implications.
2. Section 4.1.4 now documents fallback/ensemble/quantile inference resolution behavior.
3. Section 2.7 now documents frontend live-ops controls and command-linked audit navigation.
4. Baseline policy definitions and impact-table numerator/denominator mapping are now explicit for B1/B2/B3.